

RE-FUSED-6 — final report

Status: research program complete (M1 → M3/M5 → M4 → M6 → M8 → M7 → theory → M5b/M5c/M7b/M9/M9b/M10). Every number below is taken from `results/refused6/ALL_STATS.md` (generated from the CSVs) or from the verdict tables in `docs/08_defect_lineage.md`. Lineage: `docs/08`. Theory: `docs/09`. Reviewer audits: `docs/10`. Reproducibility: `docs/11`.

1. The finding in one paragraph

Gated residual correction — forecast = baseline + $g \cdot \hat{r}$, with a learned corrector $\hat{r}$ and a gate g — is usually built with the most adaptive gate available (a per-instance or neural gate). Across India daily generation, four ETT datasets and NYISO prices, three test years, four gate estimators and a tuned neural gate, **the most adaptive gate is almost never the best one**: a coarse least-squares gate (one per series, or per series × volatility regime) is as good or better, and the finer gates lose because they re-solve the residual regression on the small validation set (Proposition 1). The same price of adaptivity appears one level up: **choosing the gate granularity from validation data does not beat fixing it in advance** (pre-registered, fresh test years), and **per-instance safety mechanisms lose to a series-level deployment test**. What remains is a simple, defensible procedure and a mechanism-level account of why the fashionable alternatives fail. It is a strong applied/empirical contribution; it is not a new general-purpose ML method.

2. Final procedure (what RE-FUSED-6 recommends)

The SCALE design in `docs/07` ("Selective Correction with Adaptivity-Level Estimation") assumed the adaptivity level could be estimated from validation data. That assumption was tested and did not hold on real data (M5c, M10b), so the name is retired and the selection step removed.

```
Input : baseline B; features x; train / validation / test in time order; series id s; blocks of time b;
        alpha = 0.10; clip interval [0, 1.5]
Stage 1  corrector (train only)
         r_hat <- gradient-boosted regressor x -> R = Y - B
Stage 2  reparameterise (validation) [D9]
         g1 <- <R, r_hat> / <r_hat, r_hat> on all validation data; r_hat <- g1 * r_hat
Stage 3  per-series gate (validation) [D5, D6]
         for each series s with >= 30 validation points:
           g_s <- clip( <R_s, r_hat_s> / <r_hat_s, r_hat_s>, 0, 1.5 )
           otherwise g_s <- clip(g1 recomputed on rescaled r_hat, 0, 1.5)
Stage 4  deployment test per series (validation blocks) [Prop 7]
         skill_{s,b} <- 1 - MSE(R - g_s r_hat on block b) / MSE(R on block b)
         LCB_s <- mean_b skill - t_{1-alpha, K-1} * sd_b skill / sqrt(K)
         deploy_s <- 1[LCB_s > 0]
Output  Y_hat = B + deploy_s * g_s * r_hat(x)
Not included, because each lost when tested:
         per-instance / neural gates (M1, M1b, M8, M9, M9b, M10); shrinkage toward them (M10: period-
         dependent);
         validation-based rung selection (M5c, M10b); instance-level clipping or uncertainty filtering (M7,
         M7b)
```

Cost: one corrector fit, one scalar per series, one t-bound per series. Nothing is tuned on test data.

Per-series or per-cell? Both sit on the plateau. Per-series is the minimax-regret fixed policy on India across three test years (worst regret 0.0117, per-cell 0.0127) and on M5c real data overall (0.0020 vs 0.0026). On the hourly ETT series, with ~580 validation points per cell, per-cell was better (M5c ETT regret 0.0013 vs 0.0055; best rung in 6/12 M8 configs). The project does **not** establish a rule for choosing between them. Picking one from validation is exactly the selection step that did not pay (M5c, M10b). Use per-series as the default. Per-cell is a defensible alternative when each series × regime cell has hundreds of validation points; that suggestion is untested as a rule.

3. Evidence by stage

Stage	Question	Key result	Verdict
M1 India, 15 configs × 10 seeds	where on the adaptivity ladder is the optimum?	skill: none 0, global 0.069, per-series 0.099 , per-cell 0.094, shrunk 0.086, per-instance 0.041; best rung series 10, none 4, cell 1; oracle capture 0.223; integrity 14/14	interior optimum; per-instance worst correcting rung
M1b 4 fine estimators	is the loss an estimator artifact (D8)?	vs per-series: WLS −0.0585 (0/150), WLS-bagged −0.0531 (0/150), ridge −0.0375 (1/150), raw ratio −0.0578 (0/150)	loss survives every estimator
M1c/M1d	under-scaled corrector (D9) or binding cap (D11)?	rescaling helps the fine rung +0.0207 (49/60) but it still wins 0/60; per-series best at caps 1.0, 1.5, 3.0, ∞	loss survives both
M3/M5, M5b synthetic, 370 runs	can the rung be selected?	in-sample argmin regret 0.0384 (always picks finest); nested split 0.0031 (better in 365/370), ties fixed per-cell 0.0027	selection works in the synthetic generator
M4 ETTh1 deep baselines	does correction survive strong baselines?	per-series skill: persistence +0.244 (test MAE 2.77), NHITS +0.009 (1.49), DLinear −0.032 (1.65), PatchTST −0.068 (1.46); per-series is the best correcting rung for all four	correction value collapses as the baseline strengthens
M6 NYISO matched zones/window	does finer settlement resolution create correction value?	1-h horizon, day-ahead baseline: 15-min 0.097, 5-min 0.081, 1-h 0.077; 24-h day-ahead uncorrectable (residual $R^2 \approx 0$)	RE-FUSED-5 monotone-resolution claim falsified
M8 6 datasets × 2 horizons × 5 seeds	does the ladder transfer?	fine rung never best (0/12), loses in 11/12 configs (p=0.0032), 55/60 runs; best rungs cell 6, shrunk 3, series 2, global 1; residual- R^2 "predictor" is a scale artifact (D15)	transfers; best coarse rung is dataset-dependent
M7 India deployment rules	how to protect against harmful correction?	drifting schedule: block-LCB 0.0245, 0.4 degraded vs always 0.0032, 6.8; stationary: clip90 looked best (0.114, 0.126)	block-LCB under drift; clip90 appeared best when stationary
M5c pre-registered	does a parsimony rule fix selection noise?	real data regret: fixed series 0.0020, fixed cell 0.0026, argmin 0.0158, parsimony 0.0223	H-M5c falsified (3/3 criteria)
M7b pre-registered, fresh data	do series- and instance-level protection combine?	clip90_lcb − block_lcb −0.032 (p=9e-10); clip90 − always −0.033 on fresh data; block_lcb − always −0.004 with degraded series 4 vs 10 of 320	H-M7b falsified ; clip90 falsified out of sample
M9 pre-registered, 50 runs	does the neural sigmoid gate escape the price?	refit neural − per-series −0.0054 (better 16/50, p=3e-5); beats per-series in 2/10 configs; beats matched-data per-series +0.0063	H-M9 survives
M9b pre-registered, tuned grid	is M9 an under-tuning artifact?	tuned neural − per-series −0.0046 (better 16/50, p=1.2e-3); wins 3/10 configs	H-M9b survives
M10 pre-registered, 3 test years	is the ordering a single-period artifact?	instance best 0/8, 1/8, 0/8; below best coarse 6/8, 7/8, 6/8; pooled 19/24 (p=0.0033)	H-M10 survives
M10b pre-registered, fresh years	does rung selection beat a fixed default out of sample?	fresh 2022/2023 pooled regret: fixed series 0.0095, parsimony 0.0103, argmin 0.0116; margins 0.001–0.003	H-M10b survives ; margin small

4. Claim-by-claim: RE-FUSED-5 → RE-FUSED-6

RE-FUSED-5 claim	RE-FUSED-6 status	Decisive evidence
Shrunk hierarchical gate (SGRC) is the method	retired . Per-series beat shrunk on the 2024-25 test (M1, +0.0129) but shrunk won 5/8 configs on the 2022 test (M10): neither ordering is stable	M1, M8, M10
Free per-instance gating loses to coarse gating	established and generalised : 4 estimators, reparameterised, 4 caps, 6 datasets, 3 test years, neural gate	M1b, M1c, M1d, M8, M9, M10
$g^* = \text{SNR}/(1+\text{SNR})$ shrinkage law	narrowed : valid for noise independent of x (Prop 2); for a fixed corrector the optimal gate is $m/\bar{f}$ (Prop 1)	docs/09 §0
$\rho = W/V$ identifiability criterion ranks granularities	falsified as ranking and level rule; missing bias term (D10)	M1, M1b
Block-LCB risk control	kept as the deployment rule , not claimed as novel: only rule without a large loss anywhere	M7, M7b

RE-FUSED-5 claim	RE-FUSED-6 status	Decisive evidence
CRC-style clipping is competitive on stationary baselines	falsified out of sample (−0.033 on fresh data; destroys NYISO gains)	M7b
Correction value tracks the stationarity of the baseline's bias	established , and observed directly across origins (operator schedule +0.30 in 2022, ≤ 0 in 2023)	M1, M7, M10
Finer settlement resolution raises correction value	falsified under matched conditions	M6
Context blocks add value	null result carried forward (not re-tested)	RE-FUSED-5 R2
Cross-domain transfer of the shrunk gate	failure dissolved as a protocol artifact; what transfers is "fine gates lose"	M8
(new) validation-only rung selection	works in synthetic data; does not beat a fixed coarse default on real data	M5b, M5c, M10b
(new) a validation descriptor predicts when fine gating loses	not established (scale artifact; heterogeneity suggestive only, Holm p=0.046, n=12)	M8, D15
(new) strong deep baselines leave little to correct	established on ETTh1	M4, D12

5. Pre-registered hypotheses (all criteria fixed before the run; disclosures in docs/08)

Hypothesis	Tested on	Outcome
H-M5c parsimony selection fixes selection noise	India + ETTh1/ETTh2, 95 runs	falsified (3/3)
H-M7b hybrid series + instance protection	ETTh1/2, NYISO fresh; India in-sample	falsified (criterion i)
H-M9 neural gate does not beat per-series	India + ETTh1/2, 50 runs	survives (amendment disclosed: refit variant added after a 9-run smoke test)
H-M9b tuned neural gate does not beat per-series	same 50 runs	survives
H-M10 instance rung not best in any test year	India, test 2022 / 2023 / 2024-25	survives
H-M10b rung selection does not beat fixed per-series on fresh years	India, test 2022 / 2023	survives

Two of the program's own hypotheses were falsified and are reported as such. The narrative was changed to match.

6. M9b result (tuned neural gate)

An 8-point grid (hidden width {32, 128} × weight decay {1e-5, 1e-3} × learning rate {1e-3, 3e-3}) was trained on V1 with early stopping on V2; the lowest-V2-loss configuration was refit on all validation data.

Dataset	Baseline	h	tuned neural gate	per-series
ETTh1	persistence	1	0.1779	0.2067
ETTh1	persistence	24	0.0436	0.0498
ETTh2	persistence	1	0.1524	0.1456
ETTh2	persistence	24	0.0170	0.0181
India	persistence	1	0.1441	0.1439
India	persistence	3	0.1159	0.1193
India	roll7	1	0.2019	0.2078
India	roll7	3	0.0863	0.0860
India	snaive7	1	0.3311	0.3333
India	snaive7	3	0.1273	0.1334

Mean difference -0.0046 (tuned better in 16/50 runs, Wilcoxon $p = 1.2e-3$); per-series wins 7/10 configs. Tuning narrowed the gap from -0.0054 to -0.0046 , so the result is not an under-tuning artifact. **H-M9b survives.**

7. Defects found and fixed (D1–D17)

D1–D7 RE-FUSED-5 protocol (unit ambiguity, pooled unit types, seasonal-naive $\equiv$ persistence, resolution confound, loss mismatch, asymmetric clipping, mixed aggregations) · D8 exploding ratio gate $\rightarrow$ WLS · D9 under-scaled corrector $\rightarrow$ reparameterisation · D10 p criterion missing bias · D11 binding cap $\rightarrow$ sweep · D12 corrector starvation ruled out · D13 in-sample selection optimism $\rightarrow$ nested split · D14 nested-argmin noise $\rightarrow$ parsimony (falsified) · D15 scale confound in descriptors $\rightarrow$ relative gap + Holm · D16 parsimony fallback to "no correction" with < 4 blocks (rule withdrawn) · D17 M4 MAEs quoted from the wrong window (corrected). None was fixed by looking at test results of the hypothesis it affected; each fix was re-run and re-audited.

8. Theory status

Propositions 1–8 (docs/09_theory.md) are correct and each is tied to the experiment that tests it: exact excess risk of a gate (1), shrinkage identity (2), estimation–approximation over partitions (3), unimodality conditions (4), hold-out selection bound (5), in-sample optimism (6), the limit of validation under drift (7), and when hold-out selection cannot beat a fixed default under period shift and data halving (8). All are elementary and essentially known (forecast-combination puzzle, weighted projections, Mallows/Efron, hold-out bounds). They explain the results; they are not a theoretical contribution by themselves.

9. Limitations

- Point forecasts only; no probabilistic evaluation (CRPS, interval coverage).
 - Deep baselines on one dataset (ETTh1) and only at $h = 24$.
 - India inference uses three test years and 20 series; seeds capture model randomness, not independent time.
 - The neural gate is a small MLP (grid of 8); larger architectures were not tried.
 - No finite-sample theory for boosted or neural per-instance gates with data-dependent capacity.
 - The deployment rule's gain under drift is measured in-sample on India (M7/M7b); fresh-data evidence for it is on stationary series only, where it costs 0.004 skill.
 - Multiplicity: Holm applied within M1 and M8; the headline claims in the paper should list their test family.
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10. Novelty and venue assessment (research-fit ratings, not acceptance probabilities)

What is new. (a) A systematic, falsification-driven demonstration, with pre-registered tests and a public defect lineage, that the adaptive gates used in recent residual-correction work are dominated by a per-series least-squares gate across domains, estimators, test years and a tuned neural gate. (b) The same price of adaptivity at the selection level and at the safety level. (c) A data audit of the Indian panel (price series unusable) and a de-confounded refutation of the settlement-resolution claim. **What is not new.** The mechanism (forecast-combination puzzle; Claeskens et al. 2016), the theory (Props 1–8), block-LCB deployment (high-confidence policy improvement), hold-out selection.

Venue	Fit	Why	What a reviewer will still push on
International Journal of Forecasting	A	the result is a forecasting-methodology finding in the tradition of the combination puzzle, with multi-domain evidence and pre-registration	probabilistic evaluation; more deep baselines
Applied Energy / IEEE Trans. Smart Grid	A–	energy datasets, deployment-safety result under drifting operator schedules, resolution refutation	operational relevance of the Indian target; cost analysis
Energy and AI	A–	as above, lighter ML expectations	positioning versus residual-correction papers

Venue	Fit	Why	What a reviewer will still push on
IEEE TNNLS	B	rigorous empirical ML with mechanism	theory depth; neural gate family beyond MLP
AAAI / IJCAI	B–	clean negative result with mechanism	"known phenomenon"; limited novelty
NeurIPS time-series / forecasting workshop	A	exactly the scope: honest negative results and protocol lessons	none serious
NeurIPS / ICML / ICLR main track	not supported	the central mechanism and all theory are known; the method is deliberately simple; the program's own stopping rule applies (the gap is novelty, which experiments cannot close)	—
JMLR	not supported	would need new theory (finite-sample selection over data-dependent-capacity gates under drift)	—

Recommended path. One journal paper for IJF (primary) or Applied Energy (energy-framed), with the NeurIPS time-series workshop as an early venue for the negative result. Working title: *"The price of adaptivity in gated forecast correction: coarse gates, fixed granularity, and series-level deployment tests"*.

Appendix A — consolidated statistics (generated from CSVs)

Generated 2026-09-13T08:11:14 from the result files in results/refused6/. Nothing here is typed by hand.

Stage completion

file	present	rows
frozen_v1/M1_runs_india.csv	True	1200
M1b_runs_india.csv	True	1500
M1b_estimator_validation.csv	True	192
M1c_rescaled.csv	True	600
M1d_clip_robustness.csv	True	400
M3M5_runs_full.csv	True	370
M5b_selection_full.csv	True	370
M4_modern_baselines.csv	True	120
M6_resolution_deconfounded.csv	True	420
M8_cross_domain.csv	True	60
M7_risk_control.csv	True	270
M5c_parsimony.csv	True	280
M7b_hybrid.csv	True	350
M9_neural_gate.csv	True	300
M10_rolling_origin.csv	True	720
M10b_selection_rolling.csv	True	120
M9b_neural_tuned.csv	True	100

M1 - canonical India run

Rows: 1200 (15 configs x 10 seeds x 8 rungs). Integrity suite: 14/14 passed (frozen_v1).

Mean squared-loss skill by rung:

rung	skill
rung0_none	0.0
rung1_global	0.0693
rung2_series	0.0991
rung3_cell	0.0944
rung4_shrunk	0.0862
rung5_instance	0.0413

Best rung per config: {'rung2_series': 10, 'rung0_none': 4, 'rung3_cell': 1}

Oracle capture by best feasible rung: 0.223

M1b - stable gate estimators

rung	skill_sq_macro
rung2_series	0.0991
rung3_cell	0.0944

rung	skill_sq_macro
rung4_wls_series	0.0764
rung4_wls_cell	0.0758
rung1_global	0.0693
rung5_ridge	0.0615
rung5_wls_bagged	0.046
rung5_raw_ratio	0.0413
rung5_wls	0.0405
rung0_none	0.0

Fine estimators versus per-series (paired, unit = config x seed):

fine_estimator	minus_per_series	wins	p
rung5_wls	-0.0585	0/150	1.0e-60
rung5_wls_bagged	-0.0531	0/150	3.4e-64
rung5_ridge	-0.0375	1/150	1.8e-50
rung5_raw_ratio	-0.0578	0/150	2.3e-77

Estimator validation against the known closed-form gate:

estimator	mse_vs_gstar	corr_vs_gstar	risk	absmax
raw_ratio	0.2089	0.2814	0.8525	139996576.7028
ridge_ratio	0.1647	0.3256	0.8296	1028.8141
wls	0.1436	0.3238	0.8147	2.0692
wls_bagged	0.1328	0.3245	0.8044	1.9487

M1c / M1d - reparameterisation and cap robustness

D9 (absorb global gate into corrector):

rung	rescaled_minus_original	improves	p
rung2_series	0.002	43/60	2.7e-09
rung3_cell	0.0039	51/60	1.2e-11
rung4_shrunk	0.0087	47/60	6.8e-10
rung5_wls_bagged	0.0207	49/60	9.0e-12

D11 (admissible cap sweep) - skill by cap:

cap	rung1_global	rung2_series	rung3_cell	rung4_shrunk	rung5_wls_bagged
1.0	0.0712	0.1008	0.0996	0.0953	0.0869
1.5	0.0712	0.0942	0.0915	0.0848	0.0676
3.0	0.0712	0.0914	0.0845	0.073	0.029
inf	0.0712	0.0914	0.0842	0.0728	0.0235

M3 / M5 - synthetic mechanism and original selectors

Runs: 370 (37 configs x 10 seeds). Unimodality rate: 0.824. Mean validation-test rank correlation: 0.126

Mean regret versus oracle rung:

index	regret
regret_fixed_rung3_cell	0.0035

index	regret
regret_rho_rule	0.0037
regret_fixed_rung4_shrunk	0.005
regret_fixed_rung2_series	0.0078
regret_fixed_rung1_global	0.0082
regret_srm_c1.0	0.0306
regret_srm_c0.5	0.0371
regret_srm_c0.0	0.0381
regret_val_argmin	0.0383
regret_fixed_rung5_instance	0.0383
regret_fixed_rung0_none	0.1747

Oracle rung distribution: {'rung1_global': 124, 'rung3_cell': 117, 'rung4_shrunk': 76, 'rung2_series': 39, 'rung0_none': 12, 'rung5_raw_ratio': 2}

M5b - corrected selectors (nested split, Mallows penalty)

Runs: 370

Mean regret versus oracle rung (nested split and Mallows penalty are the corrected selectors):

index	regret
regret_fixed_rung3_cell	0.0027
regret_nested	0.0031
regret_fixed_rung4_shrunk	0.0047
regret_fixed_rung2_series	0.0078
regret_fixed_rung1_global	0.0083
regret_mallows2	0.0287
regret_mallows1	0.0373
regret_val_argmin	0.0384
regret_fixed_rung5_instance	0.0384
regret_srm_sqrt	0.0384
regret_fixed_rung0_none	0.1747

Exact oracle-rung recovery:

index	accuracy
correct_val_argmin	0.0
correct_nested	0.5027
correct_mallows1	0.0243
correct_mallows2	0.0541
correct_srm_sqrt	0.0

Regret by validation size:

n_val	regret_nested	regret_mallows1	regret_mallows2	regret_val_argmin	regret_fixed_rung3_cell	regret_fixed_rung2_series
300.0	0.0097	0.0464	0.0464	0.0464	0.0008	0.0044
600.0	0.0043	0.0626	0.0578	0.0626	0.0	0.0088
1500.0	0.001	0.049	0.04	0.049	0.0065	0.0029
3000.0	0.0024	0.0364	0.0264	0.0376	0.003	0.0083
8000.0	0.001	0.0254	0.0121	0.0254	0.0008	0.0083

n_val	regret_nested	regret_mallows1	regret_mallows2	regret_val_argmin	regret_fixed_rung3_cell	regret_fixed_rung2_series
20000.0	0.0006	0.0	0.0001	0.0082	0.0017	0.011

M5c - pre-registered parsimony selection (synthetic + fresh real data)

A_synthetic - 185 runs

index	mean_regret
regret_fixed_cell	0.0029
regret_nested_argmin	0.0031
regret_fixed_global	0.0081
regret_fixed_series	0.0082
regret_nested_parsimony	0.0117
regret_fixed_shrunk	0.0172
regret_fixed_instance	0.0392
regret_fixed_none	0.1763
comparison	mean
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nested_parsimony - nested_argmin	0.0087
nested_parsimony - fixed_series	0.0035
nested_parsimony - fixed_cell	0.0088

Tail: nested_argmin p95=0.0106 max=0.0651, nested_parsimony p95=0.0682 max=0.1893, fixed_series p95=0.0331 max=0.0797

B_real - 95 runs

index	mean_regret
regret_fixed_series	0.002
regret_fixed_cell	0.0026
regret_nested_argmin	0.0158
regret_nested_parsimony	0.0223
regret_fixed_shrunk	0.0227
regret_fixed_global	0.033
regret_fixed_instance	0.0411
regret_fixed_none	0.1254
comparison	mean
---	---
nested_parsimony - nested_argmin	0.0065
nested_parsimony - fixed_series	0.0203
nested_parsimony - fixed_cell	0.0197

Tail: nested_argmin p95=0.0736 max=0.0916, nested_parsimony p95=0.0797 max=0.0916, fixed_series p95=0.0204 max=0.0236

M4 - modern deep baselines

baseline_model	rung0_none	rung1_global	rung2_series	rung3_cell	rung4_shrunk	rung5_instance
DLinear	0.0	-0.062	-0.0324	-0.0757	-0.0597	-0.0717
NHITS	0.0	-0.0843	0.009	-0.0233	-0.0298	-0.0533

baseline_model	rung0_none	rung1_global	rung2_series	rung3_cell	rung4_shrunk	rung5_instance
PatchTST	0.0	-0.1074	-0.0678	-0.083	-0.068	-0.0893
Persistence	0.0	-0.0372	0.2442	-0.0391	0.1145	-0.0484

Baseline strength versus correction value (per-series rung):

baseline_model	base_mae	skill_sq_macro	resid_R2
DLinear	1.6503	-0.0324	0.0266
NHITS	1.492	0.009	-0.0026
PatchTST	1.4624	-0.0678	0.005
Persistence	2.766	0.2442	0.5788

M6 - de-confounded resolution study

Matched zones, window and clock horizons across resolutions.

Baseline MAE:

clock_h	baseline	15min	1d	1h	5min
1	b_da	16.6163	nan	15.5327	16.9981
1	b_pers	13.9231	nan	11.9261	14.6672
24	b_da	16.6455	10.817	15.5597	17.0282
24	b_pers	22.7833	13.9714	21.7104	23.2541

Best coarse-rung skill:

clock_h	baseline	15min	1d	1h	5min
1	b_da	0.0973	nan	0.0772	0.0807
1	b_pers	0.0394	nan	0.0913	0.0371
24	b_da	-0.0006	0.0036	-0.0001	-0.0031
24	b_pers	0.1068	0.0229	0.0947	0.1458

Out-of-sample residual R2:

clock_h	baseline	15min	1d	1h	5min
1	b_da	0.0929	nan	0.068	0.0771
1	b_pers	0.0465	nan	0.0754	0.0549
24	b_da	-0.0018	0.0028	-0.0006	-0.0056
24	b_pers	0.114	0.0213	0.0793	0.1525

M8 - cross-domain ladder and descriptors

dataset	h	resid_R2_val	resid_acf1	gate_var_between	gate_var_est	n_val_per_cell	best_coarse_skill	fine_minus_coarse
ETTh1	1	0.4091	0.2947	0.223	0.2145	580.6667	0.2087	-0.0568
ETTh1	24	0.104	0.8177	0.1285	0.53	580.6667	0.0553	-0.0123
ETTh2	1	0.1452	-0.172	0.0887	0.472	580.6667	0.1463	-0.0073
ETTh2	24	0.0269	0.8258	0.5706	1.551	580.6667	0.0185	-0.0062
ETTh1	1	0.1019	-0.0648	0.1609	0.223	2322.6667	0.0262	-0.0158
ETTh1	24	0.6436	0.9728	0.1778	0.041	2322.6667	0.3527	-0.1466
ETTh2	1	0.0441	-0.1316	0.1609	0.3019	2322.6667	0.0786	-0.0041
ETTh2	24	0.4813	0.9236	0.1042	0.0513	2322.6667	0.2595	-0.0803

dataset	h	resid_R2_val	resid_acf1	gate_var_between	gate_var_est	n_val_per_cell	best_coarse_skill	fine_minus_coarse
india_daily	1	0.2109	0.0938	0.1191	0.3254	121.6667	0.1439	-0.0104
india_daily	3	0.1849	0.6438	0.1284	0.2736	121.6667	0.1193	-0.0333
nyiso_1h	1	0.0266	-0.3007	0.0448	0.3663	2927.6667	0.0834	-0.0033
nyiso_1h	24	0.0343	0.3954	0.0282	0.2913	2927.6667	0.1029	0.0148

Best rung by dataset: {'(ETTh1', 1): 'rung3_cell', ('ETTh1', 24): 'rung3_cell', ('ETTh2', 1): 'rung4_shrunk', ('ETTh2', 24): 'rung3_cell', ('ETTh1', 1): 'rung3_cell', ('ETTh1', 24): 'rung3_cell', ('ETTh2', 1): 'rung3_cell', ('ETTh2', 24): 'rung2_series', ('india_daily', 1): 'rung4_shrunk', ('india_daily', 3): 'rung2_series', ('nyiso_1h', 1): 'rung1_global', ('nyiso_1h', 24): 'rung4_shrunk'}

Do validation-only descriptors predict the fine-rung gap? (unit = dataset x horizon)

descriptor	spearman	p
resid_R2_val	-0.8811	0.0002
gate_var_between	-0.4476	0.1446
gate_var_est	0.6853	0.0139
n_val_per_cell	0.3203	0.3102
resid_acf1	-0.5804	0.0479
base_mae_over_sd	-0.1678	0.6021

M7 - risk-controlled deployment versus alternatives

baseline	rule	coverage	degraded_series	mean_skill	withheld	withheld_helpful	worst_series_skill
operator_schedule	always	0.8367	6.8	0.0032	0.0	0.0	-0.4459
operator_schedule	block_lcb	0.18	0.4	0.0245	16.4	6.7333	-0.0472
operator_schedule	clip90	0.8367	6.6667	-0.0015	0.0	0.0	-0.4371
operator_schedule	crc_quantile	0.0033	0.0	0.0007	16.6667	9.8667	0.0
operator_schedule	never	0.0	0.0	0.0	0.0	0.0	0.0
operator_schedule	unc_selective	0.7959	6.7333	0.0045	0.0	0.0	-0.4283
persistence	always	0.9933	1.6667	0.1097	0.0	0.0	-0.0202
persistence	block_lcb	0.7233	0.8667	0.0929	5.5333	4.6	-0.0186
persistence	clip90	0.9933	0.8	0.1139	0.0	0.0	-0.0053
persistence	crc_quantile	0.0467	0.0	0.0099	18.9333	17.2667	0.0
persistence	never	0.0	0.0	0.0	0.0	0.0	0.0
persistence	unc_selective	0.9776	1.6667	0.1083	0.0	0.0	-0.0211
roll7	always	1.0	2.6667	0.1234	0.0	0.0	-0.064
roll7	block_lcb	0.6733	1.0667	0.0941	6.5333	4.9333	-0.0345
roll7	clip90	1.0	1.6667	0.1263	0.0	0.0	-0.0295
roll7	crc_quantile	0.0767	0.0	0.0189	18.4667	15.8	0.0
roll7	never	0.0	0.0	0.0	0.0	0.0	0.0
roll7	unc_selective	0.9811	3.0	0.1206	0.0	0.0	-0.0628

Block-LCB versus each alternative (paired, unit = config x seed):

rule	block_lcb_minus_rule	lcb_better	p
always	-0.0083	15/45	3.7e-02
never	0.0705	45/45	1.1e-09
crc_quantile	0.0606	45/45	2.0e-11
clip90	-0.0091	19/45	6.2e-02

rule	block_lcb_minus_rule	lcb_better	p
unc_selective	-0.0073	19/45	4.7e-02

M7b - pre-registered hybrid deployment rule

in_sample_india - 126 rule-runs

rule	degraded_series	mean_skill	withheld	worst_series_skill
always	3.0	0.0932	0.0	-0.198
block_lcb	0.4444	0.0935	8.1111	-0.028
clip90	2.7222	0.0918	0.0	-0.1808
clip90_lcb	0.1667	0.0932	8.1111	-0.0215
crc_quantile	0.0	0.0162	18.6667	0.0
never	0.0	0.0	0.0	0.0
unc_selective	3.1111	0.0917	0.0	-0.1925

fresh_stationary - 224 rule-runs

rule	degraded_series	mean_skill	withheld	worst_series_skill
always	0.3125	0.1009	0.0	0.0153
block_lcb	0.125	0.0968	1.875	0.0139
clip90	0.1562	0.0679	0.0	0.0058
clip90_lcb	0.0	0.0647	1.875	0.0027
crc_quantile	0.0	0.0389	8.4688	0.0
never	0.0	0.0	0.0	0.0
unc_selective	0.25	0.0913	0.0	0.0121

M9 - neural sigmoid gate (reviewer objection 1.3)

dataset	baseline	h	global	neural_gate	neural_gate_refit	none	per_series	per_series_V1
ETTh1	persistence	1	0.1597	0.1604	0.1728	0.0	0.2067	0.2064
ETTh1	persistence	24	0.0518	0.0507	0.0433	0.0	0.0498	0.0488
ETTh2	persistence	1	0.1462	0.1455	0.1473	0.0	0.1456	0.1437
ETTh2	persistence	24	0.0139	0.0158	0.0165	0.0	0.0181	0.0146
india_daily	persistence	1	0.1396	0.1415	0.1443	0.0	0.1439	0.1396
india_daily	persistence	3	0.0953	0.1148	0.1188	0.0	0.1193	0.1107
india_daily	roll7	1	0.1829	0.194	0.2045	0.0	0.2078	0.1831
india_daily	roll7	3	0.0408	0.0681	0.0826	0.0	0.086	0.0586
india_daily	snaive7	1	0.3168	0.3247	0.3312	0.0	0.3333	0.3204
india_daily	snaive7	3	0.0788	0.1143	0.1281	0.0	0.1334	0.1003

comparison	mean	neural_better	wilcoxon_p
neural_gate - per_series	-0.0114	6/50	0.0
neural_gate - per_series_V1	0.0004	41/50	0.0005
neural_gate - global	0.0104	41/50	0.0
neural_gate - none	0.133	50/50	0.0
neural_gate_refit - per_series	-0.0054	16/50	0.0
neural_gate_refit - per_series_V1	0.0063	40/50	0.0008
neural_gate_refit - global	0.0164	42/50	0.0

comparison	mean	neural_better	wilcoxon_p
neural_gate_refit - none	0.1389	50/50	0.0

gate	skill_sq_macro	degraded_series	gate_sd
global	0.1226	1.54	0.0
neural_gate	0.133	1.62	0.2229
neural_gate_refit	0.1389	1.36	0.2574
none	0.0	0.0	0.0
per_series	0.1444	0.84	0.3118
per_series_V1	0.1326	1.54	0.3487

M9b - neural gate with hyperparameters selected on V2, refit on V

dataset	baseline	h	neural_gate_tuned	per_series
ETTh1	persistence	1	0.1779	0.2067
ETTh1	persistence	24	0.0436	0.0498
ETTh2	persistence	1	0.1524	0.1456
ETTh2	persistence	24	0.017	0.0181
india_daily	persistence	1	0.1441	0.1439
india_daily	persistence	3	0.1159	0.1193
india_daily	roll7	1	0.2019	0.2078
india_daily	roll7	3	0.0863	0.086
india_daily	snaive7	1	0.3311	0.3333
india_daily	snaive7	3	0.1273	0.1334

neural_gate_tuned - per_series: mean -0.0046, tuned better in 16/50, Wilcoxon p=1.19e-03

Chosen configurations (hidden/wd/lr): 128/0.001/0.003 x11, 128/1e-05/0.001 x10, 128/1e-05/0.003 x10, 32/0.001/0.003 x8, 32/1e-05/0.001 x5, 32/0.001/0.001 x2, 128/0.001/0.001 x2, 32/1e-05/0.003 x2

M10 - rolling-origin India re-evaluation (reviewer objection 2.1)

origin	baseline	h	rung0_none	rung1_global	rung2_series	rung3_cell	rung4_shrunk	rung5_instance	best
origin_2022	operator_schedule	1	0.0	0.2323	0.2685	0.2629	0.3025	0.2998	rung4_shrunk
origin_2022	operator_schedule	3	0.0	0.1597	0.1784	0.1667	0.2027	0.2009	rung4_shrunk
origin_2022	persistence	1	0.0	0.1344	0.1456	0.1375	0.1384	0.1251	rung2_series
origin_2022	persistence	3	0.0	0.0907	0.0909	0.0879	0.0962	0.0906	rung4_shrunk
origin_2022	roll7	1	0.0	0.2129	0.2252	0.2258	0.2271	0.2096	rung4_shrunk
origin_2022	roll7	3	0.0	0.0816	0.0736	0.074	0.0843	0.0785	rung4_shrunk
origin_2022	snaive7	1	0.0	0.3602	0.3695	0.3721	0.3716	0.3554	rung3_cell
origin_2022	snaive7	3	0.0	0.1227	0.1339	0.1309	0.1296	0.1149	rung2_series
origin_2023	operator_schedule	1	0.0	-0.0219	-0.0169	-0.014	0.0038	0.0039	rung5_instance
origin_2023	operator_schedule	3	0.0	-0.0574	-0.0694	-0.0658	-0.0597	-0.0599	rung0_none
origin_2023	persistence	1	0.0	0.1408	0.1445	0.1425	0.1456	0.1352	rung4_shrunk
origin_2023	persistence	3	0.0	0.1293	0.1322	0.1287	0.1307	0.1214	rung2_series
origin_2023	roll7	1	0.0	0.2525	0.2545	0.2511	0.2548	0.2434	rung4_shrunk
origin_2023	roll7	3	0.0	0.1143	0.1223	0.1163	0.1107	0.1011	rung2_series
origin_2023	snaive7	1	0.0	0.4099	0.4068	0.4077	0.414	0.4056	rung4_shrunk
origin_2023	snaive7	3	0.0	0.1617	0.1724	0.1685	0.1659	0.1538	rung2_series
origin_2024	operator_schedule	1	0.0	-0.0895	0.0042	0.0004	0.0223	0.0131	rung4_shrunk

origin	baseline	h	rung0_none	rung1_global	rung2_series	rung3_cell	rung4_shrunk	rung5_instance	best
origin_2024	operator_schedule	3	0.0	-0.0521	-0.0041	-0.0055	0.0031	0.0001	rung4_shrunk
origin_2024	persistence	1	0.0	0.1396	0.1439	0.141	0.1408	0.1299	rung2_series
origin_2024	persistence	3	0.0	0.0953	0.1193	0.1172	0.1036	0.0856	rung2_series
origin_2024	roll7	1	0.0	0.1829	0.2078	0.2067	0.1906	0.1638	rung2_series
origin_2024	roll7	3	0.0	0.0408	0.086	0.0837	0.067	0.0448	rung2_series
origin_2024	snaive7	1	0.0	0.3168	0.3333	0.334	0.3224	0.2981	rung3_cell
origin_2024	snaive7	3	0.0	0.0788	0.1334	0.1307	0.0975	0.0661	rung2_series

origin	configs	instance_best	instance_below_best_coarse	mean_gap	best_counts
origin_2022	8	0	6	-0.0027	rung4_shrunk=5; rung2_series=2; rung3_cell=1
origin_2023	8	1	7	-0.0075	rung2_series=3; rung4_shrunk=3; rung0_none=1; rung5_instance=1
origin_2024	8	0	6	-0.0279	rung2_series=5; rung4_shrunk=2; rung3_cell=1

M10b - rung selection vs fixed defaults by origin (origin_2024 overlaps M5c Part B: in-sample)

origin	nested_argmin	nested_parsimony	fixed_series	fixed_cell	fixed_shrunk	fixed_global	fixed_instance	fixed_none
origin_2022	0.0148	0.014	0.0117	0.0127	0.0035	0.0244	0.0152	0.2045
origin_2023	0.0084	0.0066	0.0072	0.0078	0.0107	0.0103	0.0209	0.1706
origin_2024	0.0142	0.0237	0.002	0.0033	0.0259	0.0421	0.0509	0.1341

origin	comparison	mean	selector_lower	wilcoxon_p
origin_2022	nested_argmin - fixed_series	0.0031	10/40	0.0332
origin_2022	nested_parsimony - fixed_series	0.0023	4/40	0.0015
origin_2023	nested_argmin - fixed_series	0.0011	11/40	0.6143
origin_2023	nested_parsimony - fixed_series	-0.0007	12/40	0.9808
origin_2024	nested_argmin - fixed_series	0.0122	1/40	0.0043
origin_2024	nested_parsimony - fixed_series	0.0217	0/40	0.0

Appendix B — defect and claim lineage

Every row is a discovered problem, what it invalidated, the fix, and what the fix changed. Claims are graded: **established** / **conditionally established** / **weakened** / **reversed** / **falsified** / **open**. Nothing is deleted; superseded results stay on the record.

Defects

ID	Discovered	Problem	Invalidated	Fix	Outcome
D1	RE-FUSED-5 audit	experimental-unit ambiguity (28 cells vs 140 runs reported interchangeably)	presentation of S1	canonical unit = one (config, seed) run, declared in <code>M1_canonical_definitions.json</code>	resolved in M1
D2	RE-FUSED-5 audit	criterion test pooled 72 per-seed synthetic runs with 16 per-config real means	the central R7 statistic	aggregation only at analysis time, never across unit types	resolved in M1
D3	RE-FUSED-5 audit	seasonal-naïve at $h=7$ is literally persistence ($\text{shift}(-h+7)=\text{shift}(0)$)	4 of 12 R5 "replications"	baseline refused when $h \equiv 0 \pmod{\text{season}}$; integrity check C5 enforces it	resolved in M1
D4	RE-FUSED-5 audit	resolution study confounded (5-min used 5 zones from 2022; hourly/daily all 15 zones, full span)	the monotone-resolution claim	resolution sweep deferred to M6 with matched zones/windows/clock-horizons	deferred to M6
D5	RE-FUSED-5 audit	gates fitted under squared loss, scored on MAE	every real-data skill number	both losses computed per run; squared loss primary	resolved in M1
D6	RE-FUSED-5 audit	asymmetric clipping (fine rungs clipped, coarse rungs not)	all rung comparisons	identical $[0, 1.5]$ clip for every rung; clipped fraction logged per rung	resolved in M1
D7	RE-FUSED-5 audit	two aggregations (macro 0.0674 vs pooled 0.0902) reported side by side	cross-section comparability	both logged per run, never mixed	resolved in M1
D8	M1 audit	per-instance gate num/den explodes as $\text{den} \rightarrow 0$: measured gate variance 1e18–1e19 , 39.9% of values inadmissible, gate abs-max $\sim 1\text{e}9$	every fine-rung result in RE-FUSED-5 and M1 — "price of adaptivity" was confounded with a broken estimator	weighted-least-squares gate: regress $z = R/\hat{r}$ on x with weights $\hat{r}^2$ (exactly the projection defining g^*), so small- $\hat{r}$ points are damped analytically rather than clipped	M1b: p now 0.97–1.42 (physical); inadmissible 47.7%→24.6%; fine rung improves by ~ 0.010 but still loses to per-series → the price-of-adaptivity conclusion survives the fix
D9	M1b	$\sim 20\text{--}25\%$ of WLS gate values still exceed 1.5, i.e. the <i>corrector</i> is systematically under-scaled (l2-regularised HGB shrinks residual magnitude), so all gate estimators operate off-centre	fine-rung comparisons remain mildly handicapped	reparameterise: absorb the global gate into the corrector ($\hat{r} \leftarrow g_1 \hat{r}$) before fitting finer rungs, so admissible gates centre near 1	open — to be tested
D10	M1b	$p = W/V$ is not a valid identifiability criterion: it compares between-point variation with <i>bootstrap</i> variance and omits the estimator's bias relative to g^* . With the stable estimator $p > 1$ in most configs while the fine rung still loses	the p crossing rule as both a ranking rule (falsified in M1) and a level rule (falsified in M1b)	replace the variance-ratio diagnostic with penalised validation risk (SRM), which measures achieved risk including bias; p is retained only as a descriptive diagnostic	theory correction — carried into M5

Claim lineage

Claim	RE-FUSED-5 status	RE-FUSED-6 status after M1/M1b	Evidence
Gate identity g^* = $SNR/(1+SNR)$ and degradation condition	established (synthetic)	established , unchanged	S1: corr 0.9915 over 28 cells × 5 seeds; sign correct 25/28 cells
Free per-instance gating loses to coarse gating	established (4 settings)	established and strengthened — survives the D8 estimator fix, and the gap widens with horizon	M1: −0.058 vs per-series, 15/15 configs significant (Holm); M1b: WLS fine rung still loses in every config measured
Shrunk hierarchical gate beats the coarse gate	established (India, $p=0.0029$ synthetic)	REVERSED	M1: per-series − shrunk = +0.0129, 14/15 configs, 12 significant; sign holds under all four metric/aggregation variants
p predicts which granularity wins	falsified as ranking rule	falsified as ranking and as level rule ; cause diagnosed (D10: missing bias term)	M1 Spearman −0.561 ($p=0.030$); M1b $p>1$ while fine rung loses
The ladder has an interior optimum	new in RE-FUSED-6	established on real data	M1: per-series best in 10/15, no-correction best in the 4 drifting configs, per-cell 1/15, per-instance never
Coarse beats per-cell	new	weakened — a plateau, not a peak	block bootstrap: CI excludes zero in only 3/15 configs
Oracle headroom	new	established : best feasible rung captures 22.3% of the oracle gate	M1 audit section A
Heterogeneity moves the oracle rung finer	new	established in controlled synthetic conditions	M3 smoke: oracle rung = global at between=0.0, per-cell at between=0.3
Correcting a drifting baseline	established (RE-FUSED-5, ungated −0.29)	established and refined : with a drifting baseline every rung is negative and no-correction is optimal; coarse rungs are least harmful	M1 operator-schedule configs (4/4)
Context blocks add value	falsified (permutation control)	unchanged, carried forward	RE-FUSED-5 R2
Cross-domain transfer of the shrunk gate	falsified (ETT $p=0.80$)	unchanged; to be re-tested with the stable estimator in M8	RE-FUSED-5 R8
Resolution value rises as resolution refines	claimed	withdrawn pending M6 (confounded by D4)	—

M1b verdicts (appended after the independent estimator audit)

D8 is resolved, and resolving it strengthened rather than weakened the central claim.

Estimator validation against a closed-form target $g^* = m^2/(m^2 + \sigma_\epsilon^2)$, 6 settings × 8 seeds = 48 runs, paired by (setting, seed):

estimator	risk improvement vs raw ratio	wins	p	gate abs-max on real data
ridge ratio	+0.0229	48/48	2.9e-10	1.5e3
WLS	+0.0378	46/48	5.8e-08	4.8
WLS bagged	+0.0481	48/48	2.6e-10	4.3
raw ratio (M1 form)	—	—	—	8.6e10

Correlation with the true gate remains low throughout (0.11 at $\sigma_\epsilon=0.3$ to 0.47 at $\sigma_\epsilon=1.5$), and MSE against g^* is comparable to the magnitude of g^* itself. **The per-point gate is only weakly identifiable even in clean synthetic structure with 4,000 validation points.**

Real-data consequence (150 paired config × seed comparisons per estimator):

fine estimator	minus per-series	wins	p
WLS	-0.0585	0/150	1e-60
ridge ratio	-0.0376	1/150	1.8e-50
WLS bagged	-0.0531	0/150	3.4e-64
raw ratio	-0.0578	0/150	2.3e-77

Best rung per config is unchanged from M1 (per-series 10, none 4, per-cell 1).

Claim	Status after M1b
Adaptivity in the gate carries a statistical price	established (was conditionally established): holds for all four estimators, including the one that is provably best against a known target
Mechanism: the gate is a second-order functional of the same data that produced the corrector, so its per-point identifiability is intrinsically poor	established by direct measurement against a known g*
$\rho = W/V$ decides gate granularity	falsified at both level and ranking : with the stable estimator $\rho > 1$ in 13/15 configs while the fine rung wins 0/15; Spearman -0.389 ($p=0.15$), wrong sign
Shrinkage toward a coarse gate recovers the loss	no : rung4 variants (0.0764, 0.0758) sit well below per-series (0.0991) and per-cell (0.0944)
Reproducibility of shared rungs across independent scripts	established : M1 vs M1b max abs difference 0.00e+00

D9 status: open, under test in M1c. Fine-rung clipping remains 21–23% because the corrector under-predicts residual magnitude; the test absorbs the global gate into the corrector so that the admissible band is centred, giving the fine rungs their fairest possible test.

M1c / M1d / M3-M5 / M4 / M5b verdicts (appended 13 Sep 2026, overnight loop)

ID	Discovered	Problem	Fix	Outcome
D9	M1b	corrector under-scaled, gates off-centre	absorb global gate into corrector	resolved (M1c): fine rung +0.0207 (49/60, $p=9e-12$), gap to per-series narrows 36%, fine still wins 0/60
D11	M1c	fixed cap [0,1.5] binding after reparameterisation	cap sweep {1.0,1.5,3.0,inf}	resolved (M1d): per-series best at every cap; fine deficit widens -0.014 - > -0.068 as cap loosens (clipping was <i>helping</i> the fine rung)
D12	M4	corrector possibly data-starved vs deep baselines	internal control: persistence on identical rows and budget	resolved : persistence +0.244 with the same budget, so starvation does not explain the deep-baseline result
D13	M3/M5 audit	in-sample selection optimism : gates fitted and scored on the same validation data; val-test rank corr 0.126; argmin always chose the finest rung (regret identical to fixed rung5, 0.03839)	nested split: fit gates on V1, select on V2	resolved (M5b): regret 0.0384 -> 0.0031 (12.5x), exact recovery 0% -> 50.3%, better in 365/370 runs ($p=1e-151$)
D14	M5b	nested argmin ignores its own noise: loses to fixed coarse default at $n_{val}=300$ (+0.0089, $p=0.001$); worst-case regret 0.065 vs 0.019	parsimony (one-standard-error) rule on time blocks, $\alpha=0.10$ fixed a priori	open - pre-registered hypothesis H-M5c, tested on fresh real data

Claim	Status
Validation-only rung selection can work	conditionally established : nested split ties the hindsight-best fixed policy overall (+0.0004, $p=0.36$), beats fixed per-series ($p=2.5e-11$), and is consistent (regret 0.0097 at $n=300$ -> 0.0006 at $n=20000$). Conditional on validation size and on the heavy-tail caveat (D14).
Proposition-1 plug-in (Mallows) penalty corrects selection optimism	falsified as stated : regret 0.037 (x1) / 0.029 (x2); x2 better than x1 means true optimism exceeds twice the first-order term - the per-instance gate's effective degrees of freedom far exceed the local-cell approximation
Correction value falls as the baseline strengthens	established (M4): per-series skill +0.244 persistence (test-portion MAE 2.77) vs -0.032 DLinear (1.65), +0.009 NHITS (1.49), -0.068 PatchTST (1.46) on ETTh1 [D17: MAEs corrected to the scored test portion; whole-CV-window MAEs are 2.45 / 1.58 / 1.43 / 1.40, same ordering]; per-series is the best rung for all four baselines

Claim	Status
Resolution value rises monotonically as resolution refines	falsified (M6, matched zones/window/clock horizon): at a 1-hour horizon 15-min scores 0.097 vs 5-min 0.081 (day-ahead baseline); RE-FUSED-5's +0.445 shrinks to +0.081 once matched. What survives: correction recovers more from the weaker baseline, and the day-ahead price is uncorrectable at its own 24 h horizon (residual R2 -0.006)

M8 verdicts (appended 13 Sep 2026, 01:30)

ID	Discovered	Problem	Fix	Outcome
D15	M8 audit	absolute skill gaps scale with how predictable a dataset is: resid_R2_val correlates +0.825 with best coarse skill (p=0.001), so raw descriptor correlations with the fine-rung gap are partly mechanical	test descriptors against the RELATIVE gap (fine - coarse)/coarse and by rank partial correlation controlling best coarse skill, with Holm correction over 7 descriptors	resolved : the headline resid_R2 correlation (abs rho -0.881) does not survive on the relative scale (rho -0.427, p=0.17, Holm 0.67)

Claim	Status after M8 (6 datasets x 2 horizons x 5 seeds, persistence baseline, D8/D9 fixes)
Per-instance gating loses to the best coarse rung across domains	established : loses in 11/12 configs (binomial p=0.0032) and 55/60 runs; never the best rung (0/12). Best rungs: per-cell 6, shrunk 3, per-series 2, global 1
RE-FUSED-5's "ETT transfer failure"	dissolved : under the canonical protocol ETT behaves like India - the coarse or intermediate rung wins in 8/8 ETT configs. The specific best coarse rung (series vs cell vs shrunk) remains dataset-dependent and plateau-like
Residual predictability predicts when fine gating loses	not established - a scale artifact (D15)
Between-series gate heterogeneity predicts the proportional fine-rung loss	suggestive only : relative-gap rho -0.734 (Holm p=0.046), partial rho -0.685 (Holm p=0.12), n=12. Mechanistic reading: heterogeneity aligned with series is captured cheaply by a coarse per-series gate, leaving per-instance adaptivity proportionally less to recover
Validation size per cell predicts the gap at dataset level	not supported (relative rho +0.22, p=0.50) - confounded with dataset identity at only 3 levels; the one exception (NYISO h=24, fine wins +0.015) has the largest n per cell (2,928), consistent with but not proof of the estimation-cost mechanism

M7 verdicts (appended 13 Sep 2026, 01:35)

India daily; 3 baselines x h in {1,3,7} x 5 seeds; per-series gate with D9 reparameterisation.

Claim	Status
Block-LCB is the best deployment rule	conditionally established - baseline-dependent . Drifting operator schedule: best skill (0.0245 vs 0.0032 always) AND best safety (0.4 degraded series vs 6.8; worst -0.047 vs -0.446). Stationary persistence/roll7: dominated by CRC-style clip90 (0.1139 vs 0.0929; 0.1263 vs 0.0941) and too conservative (withholds 5.5-6.5 series, 4.6-4.9 of which would have helped)
Block-LCB is a novel safety principle	no - deploy iff a lower confidence bound on improvement is positive is standard high-confidence safe policy improvement / selective deployment. Retained as a correct tool, not a contribution
Per-instance safety mechanisms (clipping, epistemic-uncertainty selection) protect against a drifting baseline	falsified : clip90 -0.0015 (6.7 degraded), unc_selective 0.0045 (6.7 degraded), worst cases -0.43 to -0.44 - consistent with Proposition 7(iii): drift is series- and time-level, invisible to per-instance uncertainty
Conformal-quantile (worst-block) deployment	safe but nearly useless : 0 degraded everywhere, coverage 0.3-7.7%, withholds 16.7-18.9 series
Pooled over baselines	block-LCB vs always -0.0083 (p=0.037) with degraded 0.78 vs 3.71; vs clip90 -0.0091 (p=0.06) with 0.78 vs 3.04

Pre-registered hypothesis H-M7b (registered 01:35, before any result)

Series-level block-LCB and instance-level clipping address different failure modes (Proposition 7 vs Proposition 1), so the hybrid **clip90_lcb** (deploy per series iff block-LCB > 0; when deployed, clip at the validation 90th percentile) should combine them. Criteria: (i) FRESH stationary data (ETTh1, ETTh2, NYISO hourly): clip90_lcb mean skill >= block_lcb, paired Wilcoxon not significantly worse (p>=0.05 for block_lcb > clip90_lcb). (ii) FRESH stationary data: clip90_lcb degraded series

<= clip90 degraded series. (iii) India operator schedule (IN-SAMPLE, labelled as such): clip90_lcb degraded <= block_lcb degraded + 0.5. Falsified if (i) fails. alpha = 0.10 and the 90th percentile are fixed from M7, not re-tuned.

Pre-registered hypotheses H-M9 and H-M10 (registered 13 Sep 2026 ~01:38, before post_chain3 was launched at 01:39; from reviewer audit round 1)

Disclosure: a one-run smoke test of each script (India persistence h=1 seed 0 for M9; India roll7 h=1 seed 0 at the three origins for M10) was executed to check that the code runs before these criteria were written. Criteria were fixed from the audit objections, not from those numbers, and nothing was tuned afterwards.

H-M9 (reviewer 1.3 - neural gate family). Coarse dominance transfers to the field's default gate, g = 1.5*sigmoid(MLP(features, series one-hot)) trained on V1 with early stopping on V2. Runs: India {persistence, roll7, snaive7} x h in {1,3}; ETTh1, ETTh2 persistence x h in {1,24}; 5 seeds. (i) neural gate NOT significantly better than per-series over all runs (one-sided Wilcoxon p >= 0.05); (ii) neural gate beats per-series in at most half of the configs (mean over seeds). Falsified if either fails -> the dominance claim is restricted to boosted/ratio gates. per_series_V1 (per-series on V1 only) is reported as the matched-data control but is not part of the verdict.

H-M10 (reviewer 2.1 - single test period). The India ladder ordering is a property of the method, not of the 2024-25 test period. Origins test 2022 / 2023 / 2024-25, 4 baselines x h in {1,3} x 5 seeds, identical protocol (D8 WLS bagged instance gate, D9 reparameterisation, clip [0,1.5]). For EVERY origin: (i) the per-instance rung is the best rung in at most 1 of 8 configs; (ii) the per-instance rung is below the best coarse rung (global/series/cell) in more than half of the configs. Falsified if any origin fails either criterion.

Amendment to H-M9 (~01:50, before the full run started at 02:02; disclosed). A second smoke test (India persistence/roll7/snaive7, h=1, seeds 0-2 - 9 of the 50 planned runs) showed a data-budget asymmetry in the competitor: the early-stopped neural gate trains on V1 only, while per-series uses all of V (it beat the matched per_series_V1 in 9/9 runs). To give the competitor its strongest standard form, **neural_gate_refit** was added: the epoch count chosen by early stopping on V1/V2 is re-used to retrain on all of V. The H-M9 verdict is taken against whichever neural variant has the higher mean skill over all runs - a change that can only make H-M9 harder to pass. The smoke numbers (refit vs per-series -0.0023, 5/9 runs, p=0.82) come from runs that are also part of the full run; the full-run verdict is therefore not fully independent of them, and is labelled as such.

M5c verdict (appended 13 Sep 2026, ~01:58) - H-M5c FALSIFIED

Real data (India 4 baselines x h in {1,2,3,7}, ETTh1/ETTh2 persistence h in {1,24}; 95 runs). All policies are scored with the SAME gates refit on the full validation set, so the comparison has no data-budget confound.

Criterion (registered 01:20)	Result	Verdict
mean regret: parsimony <= nested argmin	0.0223 vs 0.0158 (parsimony lower in 13/95, p=0.007)	FAIL
p95 regret: parsimony < nested argmin	0.0797 vs 0.0736	FAIL
not significantly worse than fixed per-series	+0.0203, lower in 6/95, p=3.7e-10	FAIL

Synthetic replication (185 runs) points the same way: parsimony 0.0117 vs argmin 0.0031 (p=2.4e-4).

Diagnosis (audit of selections, not a re-test): * The parsimony test lacks power at the available block counts (13 ETT, 27 India blocks in V2): it picks a rung COARSER than the oracle in 54% of real runs (global 46/95 times, oracle global 4/95). Under-selection is costly - fixed global regret is 0.031 on stationary India. * The nested argmin errs in both directions (27% too coarse, 32% too fine on real data). * **Both selectors lose to fixed coarse defaults on real data:** fixed per-series 0.0020, fixed per-cell 0.0026, argmin 0.0158 (vs per-series p=1.1e-7), parsimony 0.0223. The gap is largest with the drifting operator schedule (argmin 0.043, parsimony 0.058, per-series 0.003) and on ETT (argmin 0.020, per-cell 0.001).

ID	Discovered	Problem	Fix	Outcome
D16	M5c audit	parsimony rule falls back to rung0 (no correction) when V2 has < 4 blocks; happens only at synthetic n_val=300 (3 blocks), regret 0.080	not fixed: it does not touch the real-data verdict (13-27 blocks everywhere) and the rule is withdrawn	recorded; rule withdrawn

Claim	Status after M5c
A parsimony (one-SE-style) rule fixes nested-argmin selection noise	falsified (pre-registered, real data)
Validation-only rung selection can work (M5b, synthetic)	downgraded : holds in the synthetic generator; on real data both selectors are dominated by a fixed per-series or per-cell gate
Fixed per-series is a better policy than any rung selection	suggestive, NOT yet established : per-series was itself identified as the best rung on the M1 India test period and M8 used the same ETT test split, so this comparison is partly in-sample. Tested out of sample in H-M10b

Pre-registered hypothesis H-M10b (registered ~01:59, before any fresh-origin result; M10 had not started)

Extension of M10: at each origin, the M5c selection protocol (V1 = first half of the validation year, V2 = second half, weekly blocks, $\alpha = 0.10$, identical code `m5c.real_one`) is run for India 4 baselines $\times$ h in $\{1,3\} \times 5$ seeds. `origin_2024` overlaps M5c Part B and is labelled in-sample. On the FRESH origins (test 2022, test 2023): (i) at each fresh origin, neither nested argmin nor nested parsimony has significantly lower regret than the fixed per-series gate (one-sided Wilcoxon $p \geq 0.05$); (ii) pooled over fresh origins, fixed per-series mean regret $\leq$ both selectors. Falsified if any part fails $\rightarrow$ rung selection pays on fresh test years and the fixed default is not recommended. Only `origin_2024` is smoke-tested before the run.

M7b verdict (appended 13 Sep 2026, 02:03) - H-M7b FALSIFIED (criterion i)

Fresh stationary data: ETTh1, ETTh2 (persistence, h in $\{1,24\}$, 5 seeds), NYISO hourly (persistence and day-ahead, h in $\{1,24\}$, 3 seeds) = 32 runs, 320 series-evaluations. In-sample: India persistence/roll7/operator schedule, h in $\{1,3\}$, 3 seeds.

Criterion (registered 01:35)	Result	Verdict
(i) fresh: <code>clip90_lcb</code> $\geq$ <code>block_lcb</code>	-0.0321, higher in 1/32 runs, $p=9.3e-10$	FAIL
(ii) fresh: <code>clip90_lcb</code> degraded $\leq$ <code>clip90</code>	0.00 vs 0.16 per run	PASS
(iii) in-sample drift: <code>clip90_lcb</code> degraded $\leq$ <code>block_lcb</code> + 0.5	0.17 vs 0.44	PASS

Secondary analysis on the same fresh data (NOT pre-registered; every rule was fixed before the run, nothing tuned):

Comparison (fresh, 32 paired runs)	Skill difference	Degraded series (of 320)
<code>block_lcb</code> - always	-0.0041 ($p=3.9e-4$)	4 vs 10
<code>clip90</code> - always	-0.0329 ($p=2.3e-9$)	5 vs 10
<code>unc_selective</code> - always	-0.0095 ($p=0.036$)	8 vs 10
<code>block_lcb</code> - <code>clip90</code>	+0.0289 ($p=0.028$)	4 vs 5

Per dataset, `clip90` keeps 0.019 of 0.089 (NYISO day-ahead) and 0.013 of 0.099 (NYISO persistence); on ETT it costs 0.002-0.009. In-sample India: `block_lcb` vs always is +0.034 under the drifting schedule (degraded 0.5 vs 7.2, worst series -0.08 vs -0.55) and -0.011 / -0.022 on the stationary baselines.

Claim	Status after M7b
<code>clip90</code> is the best deployment rule on stationary baselines (M7, India)	falsified out of sample : on fresh data it is the worst non-trivial rule (-0.033 vs always). Mechanism: on spiky targets (NYISO prices) the largest corrections carry most of the squared-loss gain, and a validation-quantile cap removes exactly those. The M7 result was specific to India's light-tailed residuals
Series-level and instance-level protection combine (hybrid <code>clip90_lcb</code>)	falsified (pre-registered) - it inherits <code>clip90</code> 's loss
<code>block_lcb</code> as the default safety rule	supported, secondary (not pre-registered) : the only rule without a large loss anywhere - costs 0.004 skill on fresh stationary data while cutting degraded series from 10 to 4 of 320, and gains +0.034 under drift (in-sample). Still not novel (standard high-confidence deployment test)
Per-instance safety (clipping, epistemic uncertainty) is the wrong level of protection	strengthened : fails against drift (M7) AND costs skill on fresh stationary data (M7b)

M9 / M10 / M10b verdicts (appended 13 Sep 2026, ~08:00; runs finished 02:22, PC was idle, then put to sleep by the safety net at 05:25 until the session resumed)

H-M9 SURVIVES (neural sigmoid gate; 10 configs x 5 seeds = 50 paired runs)

Comparison	Mean skill diff	Neural better	Wilcoxon p
neural_gate_refit - per_series	-0.0054	16/50	3.0e-5
neural_gate (early-stopped, V1 only) - per_series	-0.0114	6/50	5.9e-10
neural_gate_refit - per_series_V1 (matched data)	+0.0063	40/50	7.6e-4
neural_gate_refit - global	+0.0164	42/50	1.2e-8

Neural refit beats per-series in 2/10 configs (ETTh2 h=1 +0.0017; India persistence h=1 +0.0004). Largest loss: ETTh1 h=1 (0.173 vs 0.207). Degraded series per run: per-series 0.84, neural refit 1.36. Reading: the neural gate learns more than a global scalar and, given matched data, more than per-series; with the full validation set it still loses to the per-series least-squares gate. Pre-registered verdict computed against the stronger variant. Caveat: fixed a-priori hyperparameters (64 hidden, lr 1e-3, wd 1e-4) -> robustness check **M9b** (tuned grid, selected on V2 only) registered below.

H-M10 SURVIVES (rolling origin, India, 3 test years x 8 configs x 5 seeds)

Origin (test year)	Instance best	Instance below best coarse	Mean gap	Best rungs
2022	0/8	6/8	-0.0027	shrunk 5, series 2, cell 1
2023	1/8	7/8	-0.0075	series 3, shrunk 3, none 1, instance 1
2024-25 (canonical)	0/8	6/8	-0.0279	series 5, shrunk 2, cell 1
Pooled: instance best in 1/24; below best coarse in 19/24 (binomial p=0.0033).				

New finding - which coarse rung wins is period-dependent (claim downgraded). In 2022 the shrunk rung wins 5/8 configs and has the lowest fixed-policy regret (0.0035 vs per-series 0.0117); in 2024-25 per-series wins and shrunk is worst among coarse policies (0.0259 vs 0.0020). The M1 statement "per-series beats shrunk, reversing RE-FUSED-5" is therefore a property of the 2024-25 test period, not a stable ordering. The operator schedule also changes character: correctable in 2022 (shrunk +0.30), uncorrectable in 2023 (every rung <= 0 at h=3), marginal in 2024-25 - the drift that Proposition 7 describes, observed directly across origins.

H-M10b SURVIVES (rung selection vs fixed per-series on fresh test years)

Origin	argmin - fixed_series	parsimony - fixed_series
2022 (fresh)	+0.0031 (selector lower 10/40, two-sided p=0.033)	+0.0023 (4/40, p=0.0015)
2023 (fresh)	+0.0011 (11/40, p=0.61)	-0.0007 (12/40, p=0.98)
2024-25 (in-sample)	+0.0122 (1/40, p=0.004)	+0.0217 (0/40, p<1e-6)
Pooled fresh mean regret: fixed_series 0.0095, parsimony 0.0103, argmin 0.0116.		
Honest reading: on fresh years selection does not beat the fixed default, but the margin (0.001-0.003) is much		
smaller than the in-sample 2024-25 margin (0.012-0.022) - part of M5c's apparent gap was in-sample flattery of the		
per-series default.		

Fixed-policy regret across the three origins (not a verdict, reported for the recommendation): | Fixed policy | 2022 | 2023 | 2024-25 | mean | worst | |---|---|---|---|---|---| | series | 0.0117 | 0.0072 | 0.0020 | 0.0070 | 0.0117 | | cell | 0.0127 | 0.0078 | 0.0033 | 0.0079 | 0.0127 | | shrunk | 0.0035 | 0.0107 | 0.0259 | 0.0134 | 0.0259 | | global | 0.0244 | 0.0103 | 0.0421 | 0.0256 |

0.0421 | instance | 0.0152 | 0.0209 | 0.0509 | 0.0290 | 0.0509 | Per-series is the minimax-regret and mean-regret fixed policy over the three test years; it is not the best policy in every year.

Pre-registered hypothesis H-M9b (registered ~08:00, before the run started at 08:01:38)

neural_gate_tuned: 8-point grid (hidden {32,128} x weight decay {1e-5,1e-3} x lr {1e-3,3e-3}), each trained on V1 with early stopping on V2; lowest V2 loss selected; retrained on all of V for its epoch count. Same 50 runs as M9. (i) tuned neural gate NOT significantly better than per-series (one-sided Wilcoxon $p \geq 0.05$); (ii) tuned neural gate beats per-series in at most half of the 10 configs. Falsified if either fails -> H-M9 is fragile to tuning and the neural-gate claim is withdrawn.

ID	Discovered	Problem	Fix	Outcome
D17	final-package consistency check against ALL_STATS.md	M4 baseline MAEs in this file (2.45/1.58/1.43/1.40) were the whole cross-validation-window values printed in results/M4.log, while the scored ladder uses the test portion (2.77/1.65/1.49/1.46 in M4_modern_baselines.csv); the source was not stated	corrected to the CSV values with both quoted; every number in the final report is taken from ALL_STATS.md or a CSV	fixed ; ordering and conclusion unchanged

M9b verdict (appended 13 Sep 2026, 08:12; run finished 08:11:10) - H-M9b SURVIVES

Tuned neural gate (8-point grid selected on V2, refit on V), 50 paired runs: neural_gate_tuned – per_series = **−0.0046** (tuned better in 16/50, two-sided Wilcoxon $p=1.2e-3$; one-sided $p[\text{neural better}]=0.999$). Configs won: 3/10 (ETTh2 $h=1 +0.0068$, India roll7 $h=3 +0.0003$, India persistence $h=1 +0.0002$); largest loss ETTh1 $h=1$ (0.178 vs 0.207). Tuning moved the neural gate from −0.0054 (fixed hyperparameters, refit) to −0.0046: the loss is not an under-tuning artifact. Selected widths were 128 in 33/50 runs and 32 in 17/50, with no single dominant configuration.

Criterion	Result	Verdict
(i) tuned neural gate not significantly better than per-series	$p[\text{neural better}]=0.999$	PASS
(ii) beats per-series in $\leq$ half of configs	3/10	PASS

Claim	Status after M9 + M9b
The price of adaptivity extends to the neural sigmoid gate family	established for a bounded MLP gate with series one-hot input, fixed or tuned on validation, on India and ETTh1/ETTh2; untested for larger architectures or end-to-end training with the corrector